

# Open-Source Battery Smart Charger Specifications

## 1. General Description

This design is intended to provide a flexible platform for providing power to an external device using 2S Li-Ion batteries or external DC Voltage source. The module provides 8.2V – 12V power to the external system continually whilst a battery pack or external DC supply is connected.

## 2. Board Overview

### 2.1. Architecture Block Diagramm

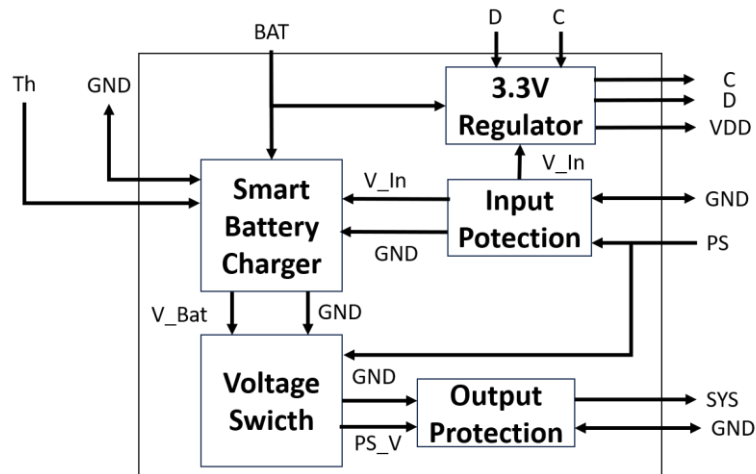


Figure 1: Architecture Block Diagramm.

### 2.2. Signal Description

| Signal Name | Description                       |
|-------------|-----------------------------------|
| PS          | Main power supply for the circuit |
| GND         | Ground reference                  |
| BAT         | Battery voltage                   |
| Th          | Analog thermistor signal          |
| SYS         | System Power Supply               |
| VDD         | 3.3V regulated power supply       |
| C           | I2C bus clock line                |
| D           | I2C bus data line                 |

Table 1: Signal Description.

### 2.3. Pin Configuration Battery Connector (J1)

| Pin No. | Symbol | Type                       |
|---------|--------|----------------------------|
| 1       | BAT    | Battery Power              |
| 2       | C      | Digital Clock              |
| 3       | D      | Digital Bidirectional Data |
| 4       | Th     | Analog Sense               |
| 5       | GND    | Ground Reference           |

Table 2: Battery Connector Pin Definition.

## 2.4. Pin Configuration System Connector (J2)

| Pin No. | Symbol | Type                         |
|---------|--------|------------------------------|
| 1       | PS     | Power Supply                 |
| 2       | GND    | Ground Reference             |
| 3       | GND    | Ground Reference             |
| 4       | D      | Digital Bidirectional Data   |
| 5       | C      | Digital Clock                |
| 6       | SYS    | External System Power Supply |

Table 3: System Connector Pin Definition.

## 3. Mechanical Specifications

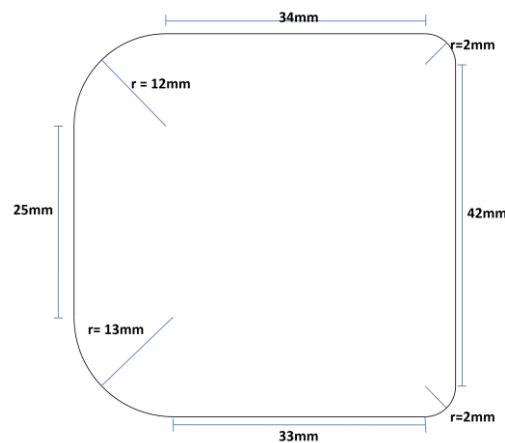


Figure 2: Board Dimensions.

## 4. Reference Schematics

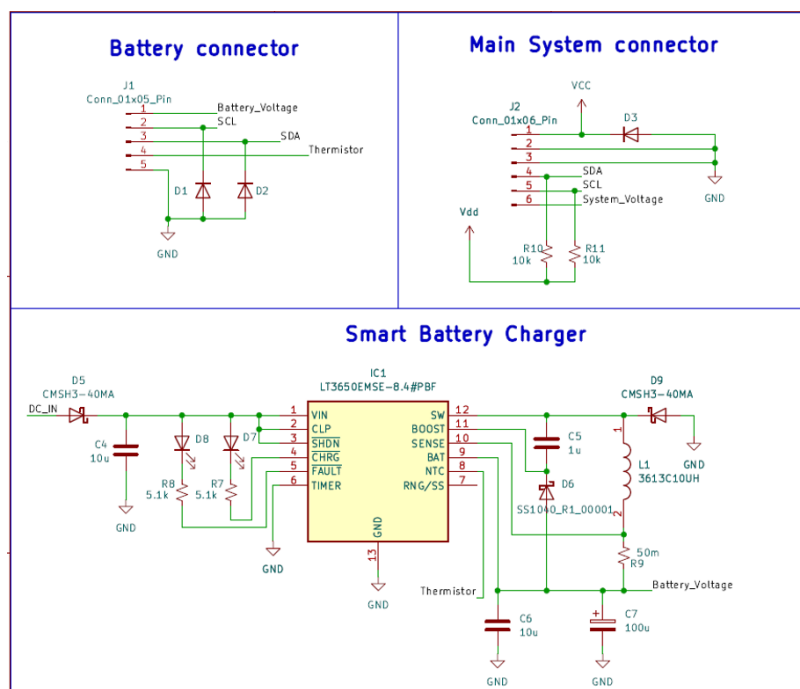


Figure 3: Connectors and Battery Charger circuitry.

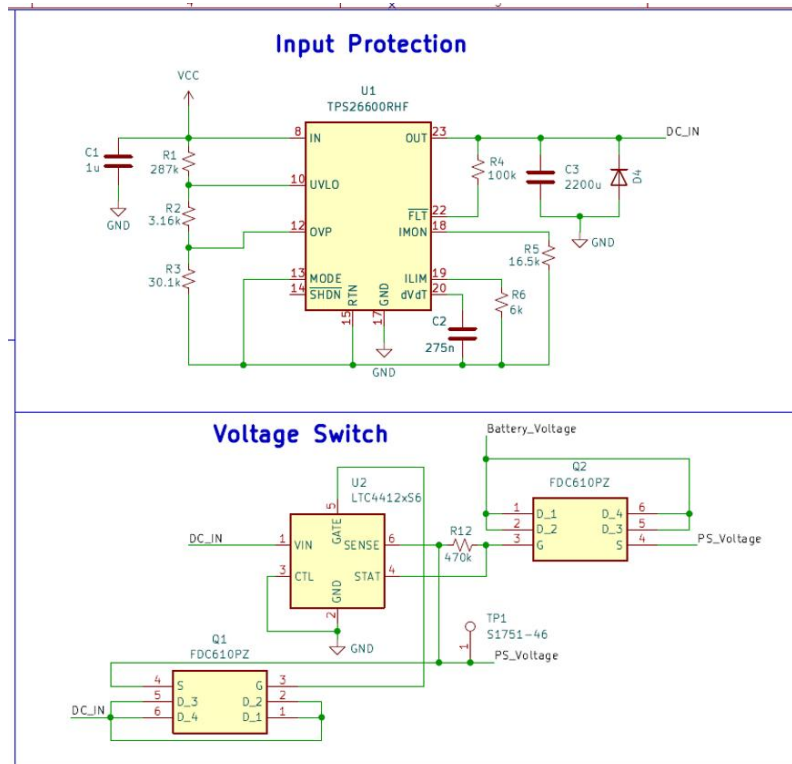


Figure 4: Input protection circuit and voltage switcher.

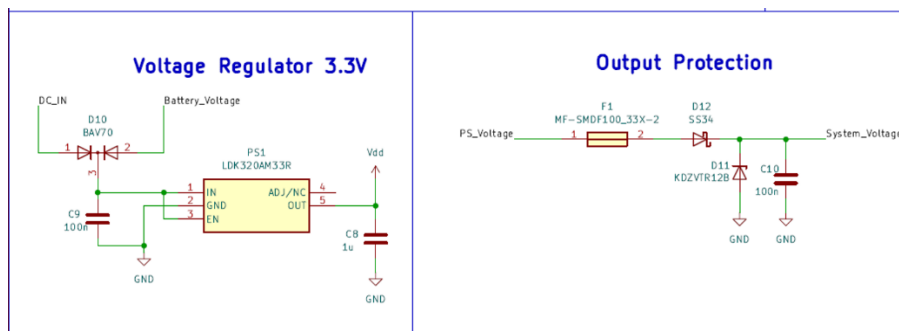


Figure 5: 3.3V Regulator and output protection circuit.

## 5. Board Specifications

### 5.1. Absolute Maximum Ratings, TA=25°C

| Parameters                    | Min.        | Max. | Unit |
|-------------------------------|-------------|------|------|
| Supply Voltage                | 12          | 12   | V    |
| Input Over Voltage Protection | -           | 12.6 | V    |
| Charging Voltage              | -           | 8.4  | V    |
| Charging Current              | -           | 2000 | mA   |
| Charge mode                   | CC/CV       |      |      |
| Reverse polarity protection   | Yes         |      |      |
| Thermistor support            | Yes         |      |      |
| Input Fuse                    | 1A polyfuse |      |      |
| Logic I2C Voltage             | -           | 3.3  | V    |

Table 4: Electrical Specifications.

## 6. Main Components Specifications

### 6.1. Input Protection (TPS26600)

The circuitry is protected from the connection from an incompatible DC Power Supply. The protection circuit has the following features:

- Undervoltage Lockout at 11.4V
- Overvoltage protection at 12.6V
- Reverse current blocking

To reach the features a TPS26600 was selected. Circuitry, shown on Figure 4 – Input Protection, is the TPS26600 configuration to protect load from overvoltage, reverse voltage, etc.

Passive components were calculated following the next procedure:

System design parameters:

$$\begin{aligned} V_{in} &= 12V \\ V_{OV} &= 12.6V \\ V_{UV} &= 11.4V \\ RL(su) &= 12\Omega \\ I_{lim} &= 2A \\ C_{Out} &= 2200\mu F \end{aligned}$$

#### *Undervoltage Lockout and Overvoltage Protection*

From Datasheet  $V_{OVPR} = 1.19V$  and  $V_{UVLOR} = 1.19V$ .

Overvoltage protection ( $V_{OV} = 12.6V$ ) and Undervoltage protection  $V_{UV} = 11.4V$ . Values come from power supply (AKM45US12). Resistor values calculated starting with  $R_3 = 30.1k\Omega$  and equations from datasheet.

$$\begin{aligned} X &= R_1 + R_2 + R_3 \\ X &= \frac{R_3}{V_{OVPR}} = 318.706 k\Omega \\ R_2 &= \frac{V_{UVLOR}}{V_{UV}} = 3.168 k\Omega \\ R_1 &= X - R_2 - R_3 = 285.437 k\Omega \end{aligned}$$

Looking for real resistor values with case code 0805 (in) available on Mouser Electronics:

$$\begin{aligned} R_1 &= 287 k\Omega \\ R_2 &= 3.16 k\Omega \\ R_3 &= 30.1 k\Omega \end{aligned}$$

Calculating new voltage values:

$$\begin{aligned} V_{OV} &= V_{OVPR} \cdot \frac{R_1 + R_2 + R_3}{R_3} = 12.661 V \\ V_{UV} &= V_{UVLOR} \cdot \frac{R_1 + R_2 + R_3}{R_2 + R_3} = 11.458 V \end{aligned}$$

#### *Active Current Limiting*

Proposed System max. current (2A) shall be the limit. The equations are from datasheet:

$$R_{ilim} = \frac{12}{I_{lim}} = 6 k\Omega$$

*Input noise suppression*

$C_{In}$  has a typical capacitance value in range of  $0.1\mu F$  to  $1\mu F$ .

$$C_{In} = 1\mu F$$

*Current Monitoring Resistor*

For  $I_{lim} = 2A$ , and considering the operation voltage range of ADC from  $0V$  to  $2.5V$ :

$$V_{IMON_{max}} = 2.5 V$$

$$R_{IMON} = \frac{V_{IMON_{max}}}{I_{lim} \cdot (75 \cdot 10^{-6})} = 16.667 k\Omega \quad V = I_{lim} \cdot (75 \cdot 10^{-6}) \cdot 16.5 k\Omega = 2.475 V$$

Real value for  $R_{IMON} = 16.5 k\Omega$

*Output Voltage Ramp Time*

For an inrush current  $I_{inrush} = 1 A$  (estimated) shall be the  $t_{dVdT}$  calculated:

$$t_{dVdT} = C_{out} \cdot \frac{V_{In}}{I_{inrush}} = 65.34 ms$$

For a given start-up time,  $C_{dVdT}$  capacitance value is calculated:

$$C_{dVdT} = \frac{t_{dVdT}}{(8 \cdot 10^3) \cdot V_{In}} = 275 nF$$

The pull-up resistor ( $R_4$  or  $R_{flt}$ ) for the open-drain fault output shall have a value in the range of  $10 k\Omega$  to  $100 k\Omega$ . The current sink by pin FLT must not exceed  $10 mA$ .

$$R_{flt} = 100 k\Omega$$

$$I_{flt} = \frac{12V}{R_{flt}} = 0.297 mA$$

## 6.2. Battery Charger (LT3650-8.4)

The battery pack can be charged from an external 12V ACDC power supply. For design the LT3650-8.4 was selected. Some features of the controller are:

- Constant-Current/constant-voltage charge
- 2A charging current
- Low battery and bad-battery detection
- Auto-restart mode if the battery voltage drops 2.5%

The LT3650 is a Li-Ion battery charger, which incorporates a 2 A switch that is driven by a bootstrapped supply to maximize efficiency during charging cycles. Once charging is terminated and the LT3650 is not actively charging, the IC automatically enters a low current standby mode in which supply bias currents are reduced to  $85 \mu A$ . The IC monitors battery temperature by using a thermistor during charging cycle, suspends charging, and signals a fault condition if the battery temperature moves outside a safe charging range.

Passive components were calculated following the next procedure:

From datasheet we got the next values

$$C_{in} = 10 \mu F \quad C_{Bat} = 10 \mu F \quad C_{Bat2} = 100 \mu F$$

The design inputs for this circuitry are:

$$I_{max} = 2 \text{ A} \quad \Delta I_{max} = 0.25 \cdot I_{max} = 0.5 \text{ A} \quad V_{Bat} = 8.2 \text{ V} \quad V_{in} = 12 \text{ V} \quad V_f = 420 \text{ mV}$$

Then the calculated values for passive components are:

$$R_{sense} = \frac{0.1 \text{ V}}{I_{max}} = 0.05 \Omega$$

$$L = \left( \frac{10}{\Delta I_{max}} \right) \cdot R_{sense} \cdot (V_{Bat} + V_f) \cdot \left( 1 - \frac{V_{Bat} + V_f}{V_{in} + V_f} \right) = 2.637 \mu\text{H}$$

The calculated value is too small for a robust system, therefore the proposed value from datasheet will be used.

$$L = 10 \mu\text{H}$$

For diode election the next value is needed:

$$I_{diode} > \frac{I_{max} \cdot (V_{in} - 5.65)}{V_{in}} \sim 1.0583 \text{ A}$$

It means the  $I_{diode} > 1.1 \text{ A}$

### 6.3. Voltage Switch (LTC4412)

The LTC4412 controls an external P-Channel MOSFET for power switchover. When a wall adapter is connected the load is automatically disconnected from the battery. For this design the primary supply is the wall adapter. The capability of the automatic controller makes it feasible to use the IC without an external microcontroller.

### 6.4. Output Protection

The fuse used protects the circuit from overcurrents and short circuits; it does not blow or suffer permanent damage but is capable of automatically resuming normal operation after cooling down. In addition, a Schottky diode is used to provide reverse current protection, along with a Zener diode and a capacitor in parallel to prevent overvoltage clamping, absorb spikes, reduce ringing, and improve transient stability.

### 6.5. Voltage Regulator

The LDK320 is a voltage regulator, which provides a 3.3V output. The very good dynamic characteristic, combined with low drop voltage and low quiescent current make it suitable for this application. This device also includes short-circuit current limiting, thermal and safe operating area protection.

## 7. Performance

To verify that this circuit is working properly, the ideal outcome would be to obtain the following results.

- Full battery charge, with graphs showing Battery Charging Current (Ampere) vs Battery Charging Time Percentage (%); Battery Charging Current (Ampere) vs Battery Voltage (Voltage); Temperature (Celsius degree) while Battery Charging (%); Battery Voltage (Voltage) while Battery charging (%); Power Supply Current (Ampere) while Battery charging (%); Power comparison (W) between circuit input (Power Supply) and output

(Battery) during charging process; Efficiency (%) vs Battery Charging Time Percentage (%).

- Battery discharge, graphically representing Temperature (°c) while Battery discharging; Battery Voltage while discharging with different loads.
- Simultaneous connection of the power supply and the fully charged battery, graphically representing efficiency.
- Battery discharge with loads varying over time, graphically representing performance.

## 8. Bill Of Materials

| Reference          | Value              |
|--------------------|--------------------|
| <b>C1,C5,C8</b>    | 1u                 |
| <b>C2</b>          | 275n               |
| <b>C3</b>          | 2200u              |
| <b>C4,C6</b>       | 10u                |
| <b>C7</b>          | 100u               |
| <b>C9,C10</b>      | 100n               |
| <b>D1,D2,D3,D4</b> | SMAJ12A-HP6AT/13   |
| <b>D5,D9</b>       | CMSH3-40MA         |
| <b>D6</b>          | SS1040_R1_00001    |
| <b>D7</b>          | Q62702P5179        |
| <b>D8</b>          | Q62702P5178        |
| <b>D10</b>         | BAV70              |
| <b>D11</b>         | KDZVTR12B          |
| <b>D12</b>         | SS34               |
| <b>F1</b>          | MF-SMDF100_33X-2   |
| <b>IC1</b>         | LT3650EMSE-8.4#PBF |
| <b>J1</b>          | Conn_01x05_Pin     |
| <b>J2</b>          | Conn_01x06_Pin     |
| <b>L1</b>          | 3613C10UH          |
| <b>PS1</b>         | LDK320AM33R        |
| <b>Q1,Q2</b>       | FDC610PZ           |
| <b>R1</b>          | 287k               |
| <b>R2</b>          | 3.16k              |
| <b>R3</b>          | 30.1k              |
| <b>R4</b>          | 100k               |
| <b>R5</b>          | 16.5k              |
| <b>R6</b>          | 6k                 |
| <b>R7,R8</b>       | 5.1k               |
| <b>R9</b>          | 50m                |
| <b>R10,R11</b>     | 10k                |
| <b>R12</b>         | 470k               |
| <b>TP1</b>         | S1751-46           |
| <b>U1</b>          | TPS26600RHF        |
| <b>U2</b>          | LTC4412xS6         |

## 9. Test Procedures

### 9.1. Functional Test

1. Do a visual inspection, verifying all connectors are well welded according to IPC-A-610E SP, all IC's are correctly oriented, if PCB has scratches or any mechanical damage, also look for solder bridges.
2. Verify that J1 pin 1 (Battery\_Voltage), J2 pin 1 (VCC) and pin 6 (System\_Voltage), IC1 pin 1 (DC\_IN), TP1 and connection between D1, D2, D3, D4, D9, D11 have non-connectivity with GND.
3. Connect J2 to AC/DC Wall Adapter 12V. D8 LED shall be OFF the whole time.
4. TP1 should have a value around 12V.
5. Connect a multimeter and an electric load at 2A to J2 (System\_Voltage and GND).
6. Turn on the electronic load and look at the output voltage in oscilloscope. Voltage must be around 11.5V.
7. Turn off the electric load and disconnect J2 and connect battery pack to J1. D7 must turn on if Power Supply and Battery are connected simultaneously.
8. Repeat procedure instructions 4 to 6. But Voltages shall be between 5.2V and 8.4V.
9. Disconnect battery.

### 9.2. Discharging Test

1. Before continuing with this test, the functional test shall be approved.
2. Be sure Battery is full.
3. During this test is important to get the next values:
  - a. Battery current in Amperes.
  - b. Battery voltage in Volts.
  - c. Power Supply current in Amperes.
  - d. Power Supply voltage in Volts.
  - e. Temperature in Celsius.
4. To acquire each value is needed to use a specific connection. Connections shall be explained only if this board is manufactured because external custom boards or harnesses would be needed.
5. Connect harness at J2.
6. Identify System\_Voltage and GND pins. Connect an electric load to them.
7. Connect the battery and set the electric load at 2A.
8. After doing the connections to measure data, turn on the electric load.
9. When battery is empty, turn off the electric load and stop data acquisition. Save the data.
10. Disconnect all and storage the equipment as specified in laboratory instructions.

### 9.3. Charging Test

1. Before continuing with this test, the functional test shall be approved.
2. Be sure Battery is empty.
3. During this test is important to get the next values:
  - a. Battery current in Amperes.
  - b. Battery voltage in Volts.
  - c. Power Supply current in Amperes.

- d. Power Supply voltage in Volts.
- e. Temperature in Celsius.
4. To acquire each value is needed to use a specific connection. Connections shall be explained only if this board is manufactured because external custom boards or harnesses would be needed.
5. Connect harness at J2.
6. Connect the battery and the Power Supply. Harness J2 will have a DC Jack connector.
7. Connect Power Supply to AC Socket.
8. When battery is full, disconnect Power Supply and stop data acquisition. Save the data.
9. Disconnect all and storage the equipment as specified in laboratory instructions.

## 10. Equipment needed

The equipment needed to run the tests to this circuit is:

- Digital Multimeter
- Power Supply or Voltage Generator
- Temperature sensor
- Electronic Load

## 11. Design Inputs

**Requirement:** The circuit should be protected from the connection from an incompatible DC Power Supply. The protection circuit should have the following features:

- Reverse Polarity Protection
- Overvoltage protection

**Actions:** TPS26600 provides over voltage protection at 12.6V since it is considered that Power Supplies normally have variations at output, having a protection exactly at 12 V will limit the supplies that can be used, while operating range is 60V and reverse input polarity protection is -60V.

**Requirement:** The battery should be charged from an external 12V ACDC power supply. A compatible barrel jack connector should be used with the DC Supply.

**Actions:** Components are configured to work with an external 12V ACDC Power Supply (AKM45US12).

**Requirement:** The board should pass the output voltage directly from the charging circuit to the device output without alteration.

**Actions:** There is no internal regulator for main voltage output. Value at input is almost the same value at the output, it depends on the characteristics from Supply with a load.

**Requirement:** SMBus pins should be exposed on the output connector. Logic levels should be 3.3V.

**Actions:** Communication logical voltage is provided by LDK320AM33R.

**Requirement:** The output of the board should be protected in such a way that failure in either the external load or the board itself should not damage the other module.

To mitigate this risk the board should provide the following output protection features.

- Overcurrent protection outside of supply specification.
- Over voltage protection outside of supply specification.

Actions: To solve an unfortunate failure as overcurrent or over voltage there is a diode (KDZVTR12B) and fuse (MF-SMDF100\_33X-2) arrangement to provide protection to module that is connected to the board.